

# Combining Technical Indicators and Equal-Budget Metaheuristic Optimization for Short-Term Mini-Index Futures Prediction

## Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound  $0.60 \text{ MCC} + 0.40 F_1$  fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

**Keywords:** Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

## 1 Introduction

Short-term price forecasting in high-frequency derivatives markets represents one of the most computationally demanding applications of neural computing [1, 2]. Among traded financial instruments, futures contracts are particularly sought after by institutional and retail market participants due to their high liquidity, leverage potential, and utility for both risk hedging against price fluctuations and speculative trading

strategies [3, 4]. These contracts establish an agreement to purchase or sell an underlying asset at a predetermined price on a future date, rendering them exceptionally sensitive to order flow dynamics, short-term market expectations, and macroeconomic announcements [5, 6]. In Brazil, the Mini-Index futures contract (symbol: WIN) on Brasil Bolsa Balcão (B3), referenced to the Ibovespa stock index, exhibits immense daily trading volume and constitutes a prime domain for computational intelligence research in financial time series [1, 7]. However, price series derived from 5-minute intraday candles exhibit severe non-stationarity, low signal-to-noise ratios, structural regime shifts, and rapid dissipation of arbitrage patterns as market participants exploit temporary inefficiencies [8, 9]. These characteristics render accurate directional trend classification a methodologically challenging task that demands both robust feature engineering and principled hyperparameter optimization [10].

Early attempts to formalize market patterns and predict directional movements relied on static linear models, such as autoregressive integrated moving average (ARIMA) techniques [11]. As finance and computer science converged under the computational intelligence paradigm [12], data-driven machine learning (ML) models gained widespread adoption for financial classification tasks. Supervised learning algorithms—including Random Forests (RF) [13], Support Vector Machines (SVM) [14], Multilayer Perceptrons (MLP) [15], and 1D Convolutional Neural Networks (1D-CNN) [16]—have demonstrated strong capabilities in learning complex, non-linear decision boundaries from historical market features. A central prerequisite for building effective classifiers involves technical analysis, wherein mathematical transformations of price and volume series—such as Simple Moving Averages (SMA), Exponential Moving Averages (EMA), and standard deviation envelopes—are computed to capture trend momentum and volatility regimes [10, 17]. Nevertheless, computing high-dimensional pools of technical indicators introduces significant redundancy and noise, creating a feature space that degrades classifier generalization and increases computational overhead [18, 19]. Consequently, applying feature selection techniques to extract compact, highly informative indicator subsets becomes vital for stabilizing training dynamics and improving out-of-sample robustness [3, 20].

Beyond feature engineering and selection, classifier performance depends critically on hyperparameter configuration [21]. In neural network architectures such as MLPs and 1D-CNNs, structural parameters (e.g., hidden layer width, filter counts), regularization coefficients (L2 weight decay  $\alpha$ , dropout rate  $p_{\text{drop}}$ ), and training settings (learning rate  $lr$ , mini-batch size  $B$ ) jointly define a complex, non-convex optimization surface over a mixed discrete-continuous search space  $\theta \in \mathbb{R}^D$  [9, 22]. Traditional gradient-based algorithms such as Stochastic Gradient Descent or Adam operate effectively on differentiable weight-loss surfaces but cannot directly navigate the non-differentiable, combinatorial hyperparameter landscape [23, 24]. To overcome these limitations, metaheuristic optimization algorithms (MOAs)—spanning evolutionary algorithms such as Genetic Algorithms (GA) [25] and Differential Evolution (DE) [26], as well as swarm intelligence paradigms such as Particle Swarm Optimization (PSO) [27] and Grey Wolf Optimizer (GWO) [23]—have gained substantial prominence for automated hyperparameter tuning in neural computing applications [1, 4]. These population-based methods maintain diverse candidate pools and employ stochastic

exploration-exploitation operators, enabling effective traversal of rugged, multi-modal fitness landscapes without requiring gradient information [23].

Despite the growing literature on metaheuristic-tuned financial classifiers published in premier neural computing journals [1, 4, 9, 16], existing financial machine learning studies overwhelmingly suffer from three major methodological flaws:

1. **The Budget Inequality Flaw:** Benchmark studies frequently evaluate metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing a GA running for 100 generations with a PSO running for 30 iterations) [9]. Because stochastic search performance scales monotonically with candidate evaluation count, reported performance superiority in prior studies is often an artifact of unequal evaluation budgets rather than inherent algorithmic efficiency.
2. **The Temporal Data Leakage Flaw:** Standard randomized  $K$ -fold cross-validation or randomized data shuffling is routinely applied to time series data [8, 9]. Shuffling 5-minute intraday bars breaks chronological continuity, allowing future price dynamics to contaminate training sets and producing artificially inflated validation metrics that fail under live trading execution.
3. **The Classification-Economic Disconnect Flaw:** Models are overwhelmingly tuned and evaluated solely on raw classification accuracy or cross-entropy loss [28]. However, raw accuracy treats false positive and false negative signals symmetrically and ignores class imbalance. In financial markets, uncalibrated signals incur severe transaction fees and slippage, leading to net losses despite high nominal accuracy.

To resolve these methodological limitations, this paper presents a controlled multi-optimizer benchmark framework for short-term neural intraday trend classification on 5-minute B3 Mini-Index futures data. Incorporating Information Gain feature selection to extract a compact 7-indicator subset from 66 raw technical indicators, we systematically evaluate five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—across four diverse classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation budget cap of  $N_{\text{eval}} = 1,500$  evaluations per seed within a leakage-free 60/20/20 chronological split, optimizing a compound MCC-driven objective ( $0.60\text{MCC} + 0.40F_1$ ) designed to resist class imbalance distortion. Through non-parametric Friedman and post-hoc Wilcoxon–Holm analyses, we demonstrate statistical competitive parity among GA, DE, and PSO under budget equality, and validate the top-ranked configuration via out-of-sample financial backtesting under transaction costs.

The remainder of this manuscript is organized as follows: Section 2 situates the study within the literature on neural optimization, feature selection, and metaheuristics in financial forecasting. Section 3 details the dataset generation, Information Gain feature selection, search-space formulations, metaheuristic operators, and evaluation design. Section 4 presents the empirical benchmark results, convergence dynamics, statistical analyses, and financial backtest evaluations. Finally, Section 5 concludes the paper and outlines future research directions.

## 2 Systematic Review of Literature

Predicting future price direction in high-frequency financial markets remains a central challenge in computational finance due to extreme signal-to-noise ratios, market regime shifts, and non-stationary distribution properties [29]. Machine learning models and technical indicator representations have been widely deployed to capture non-linear market dependencies; however, model performance depends heavily on feature subset selection, hyperparameter tuning, and validation protocol integrity. This section presents a structured Systematic Review of Literature (SRL) examining recent research across feature selection, metaheuristic optimization, and temporal validation designs, establishing the theoretical landscape and identifying critical methodological gaps.

### 2.1 Literature Selection Protocol and Research Questions

To ensure rigorous synthesis and eliminate selection bias, literature was retrieved following an explicit PRISMA-guided protocol across major scholarly databases: IEEE Xplore, Scopus, Web of Science, and SpringerLink (specifically targeting *Neural Computing and Applications*). Search queries combined boolean terms for optimization paradigms ("metaheuristic", "genetic algorithm", "particle swarm", "differential evolution", "grey wolf"), computational models ("neural network", "support vector machine", "random forest", "1D-CNN"), and financial applications ("futures market", "intraday forecasting", "technical indicators").

The review was bounded to peer-reviewed studies published between 2015 and 2026. Inclusion criteria required papers to evaluate automated feature selection or hyperparameter tuning on financial or complex time series. Studies lacking reproducible validation metrics or failing to detail search configurations were excluded. The systematic review is framed by three central Research Questions (RQs):

1. **RQ1 (Budget Parity):** How do stochastic metaheuristic search operators perform across distinct classifier search spaces when operating under a strictly equal candidate evaluation budget cap ( $N_{\text{eval}} = 1,000$ )?
2. **RQ2 (Temporal Leakage):** What is the quantitative impact of temporal data leakage when comparing chronological sequential splits against randomized cross-validation on out-of-sample trend prediction?
3. **RQ3 (Economic Alignment):** How effectively do classification-based fitness objectives translate into out-of-sample trading performance (Net Return, Sharpe ratio, Maximum Drawdown) under live exchange transaction costs?

### 2.2 Classifier Backbones and Input Representations

Supervised classification of intraday price direction relies heavily on input feature quality. Raw high-frequency bar data (OHLCV) exhibit high volatility and noise. Consequently, technical analysis indicators are widely engineered to extract momentum,

trend strength, and volatility signals [24]. However, calculating dozens of indicators across multiple lookback horizons introduces severe multicollinearity and high dimensionality.

Information-theoretic attribute evaluation, such as Information Gain (IG) filtering, provides a reproducible, non-parametric mechanism to rank indicators according to their mutual information with the target class label [19]. Feature selection mitigate the curse of dimensionality, stabilizing fitted model boundaries and accelerating downstream optimization.

Downstream predictive backbones span diverse hypothesis spaces:

- **Random Forest (RF)**: An ensemble of randomized decision trees effective at capturing non-linear feature interactions while resisting overfitting through bagging.
- **Support Vector Machines (SVM)**: Maximum-margin kernel classifiers capable of projecting non-linearly separable indicator subsets into high-dimensional Hilbert spaces.
- **Multilayer Perceptron (MLP)**: Fully-connected feedforward neural networks that learn non-linear decision functions through backpropagation and non-linear activation layers.
- **1D Convolutional Neural Network (1D-CNN)**: Deep neural architectures employing spatial and temporal 1D convolution kernels to extract local temporal patterns directly from sequential indicator windows.

Prior studies confirm that classifier ranking varies substantially across input representations, highlighting that hyperparameter configuration and model topology must be tuned jointly with input space filtering [13, 30].

## 2.3 Metaheuristic Optimization Paradigms

Hyperparameter optimization is vital to prevent underfitting or severe overfitting in financial learning models. Traditional gradient-based optimization algorithms (e.g., SGD, Adam) require differentiable loss functions and are prone to entrapment in local optima when navigating non-convex hyperparameter spaces [9]. Population-based metaheuristics offer derivative-free global search mechanisms across complex discrete-continuous search spaces.

Four primary population-based metaheuristic families are widely applied:

- **Genetic Algorithms (GA)**: Evolutionary algorithms leveraging natural selection, crossover, and mutation operators to evolve candidate solution populations [21, 25].
- **Particle Swarm Optimization (PSO)**: Swarm intelligence paradigms simulating social flocking behaviors, where particles adjust trajectories based on personal historical bests and global swarm achievements [27, 31].
- **Differential Evolution (DE)**: Stochastic population-based vector search methods utilizing vector differences for self-referential mutation and crossover operations [26].
- **Grey Wolf Optimizer (GWO)**: Bio-inspired metaheuristics modeling the hunting hierarchy ( $\alpha, \beta, \delta, \omega$ ) and prey encirclement mechanisms of grey wolves [4, 23].

Random Search (RS) serves as an indispensable baseline in metaheuristic benchmarks, providing a transparent, unbiased reference point under identical candidate evaluation budgets [32].

Despite widespread adoption, prior studies frequently compare metaheuristics under unconstrained or unequal evaluation counts (e.g., comparing 100 generations of GA against 30 iterations of PSO without matching population size), making it impossible to determine whether reported performance gains stem from superior search operators or simply greater computational budget [9, 15].

## 2.4 Comparative Synthesis of Representative Studies

To establish the state of the art and identify persistent methodological weaknesses, Table 1 synthesizes 15 representative studies from 2015 to 2026 alongside the present work. Studies are evaluated across six technical dimensions: publication context, application domain and dataset, classifier backbones, optimization and feature paradigms, validation protocols, and identified methodological gaps.

## 2.5 Critical Methodological Gaps and Research Motivation

As synthesized in Table 1, three critical methodological flaws permeate existing research:

1. *The Budget Inequality Flaw*: Prior literature routinely compares metaheuristic optimizers without capping the candidate evaluation budget ( $N_{\text{eval}}$ ). Performance claims favoring complex algorithms are often confounded by unequal function evaluations rather than superior search operator efficiency [9].
2. *The Temporal Data Leakage Flaw*: Financial forecasting studies overwhelmingly employ randomized  $K$ -fold cross-validation or data shuffling. Shuffling sequential financial bars allows future market information to contaminate training folds, producing artificially inflated accuracy scores that fail out-of-sample [8, 35].
3. *The Classification-Economic Disconnect Flaw*: Models are tuned almost exclusively on raw accuracy or cross-entropy loss, ignoring class imbalance and live trading frictions. In real-world futures markets, a classifier with high nominal accuracy can incur net financial losses due to transaction fees and uncalibrated directional signals [3, 28].

To resolve these methodological limitations, this work establishes a controlled, equal-budget benchmark framework on 15,057 five-minute B3 Mini-Index futures bars. Incorporating Information Gain feature selection to extract a compact 7-indicator subset, we systematically evaluate five distinct optimizer families (RS, GA, PSO, DE, GWO) across four classifier backbones (MLP, SVM, RF, 1D-CNN). All optimizers operate under a strictly equal candidate evaluation cap of  $N_{\text{eval}} = 1,000$  evaluations per seed across three locked common seeds within a leakage-free 60/20/20 chronological split. Furthermore, we contrast two distinct validation fitness objectives (a compound  $0.60 \text{ MCC} + 0.40 F_1$  objective vs. weighted accuracy) and evaluate out-of-sample financial profitability under exchange transaction fees.

**Table 1** Systematic Literature Review comparison matrix auditing representative studies in financial machine learning, metaheuristic hyperparameter tuning, and temporal validation protocols against identified methodological gaps.

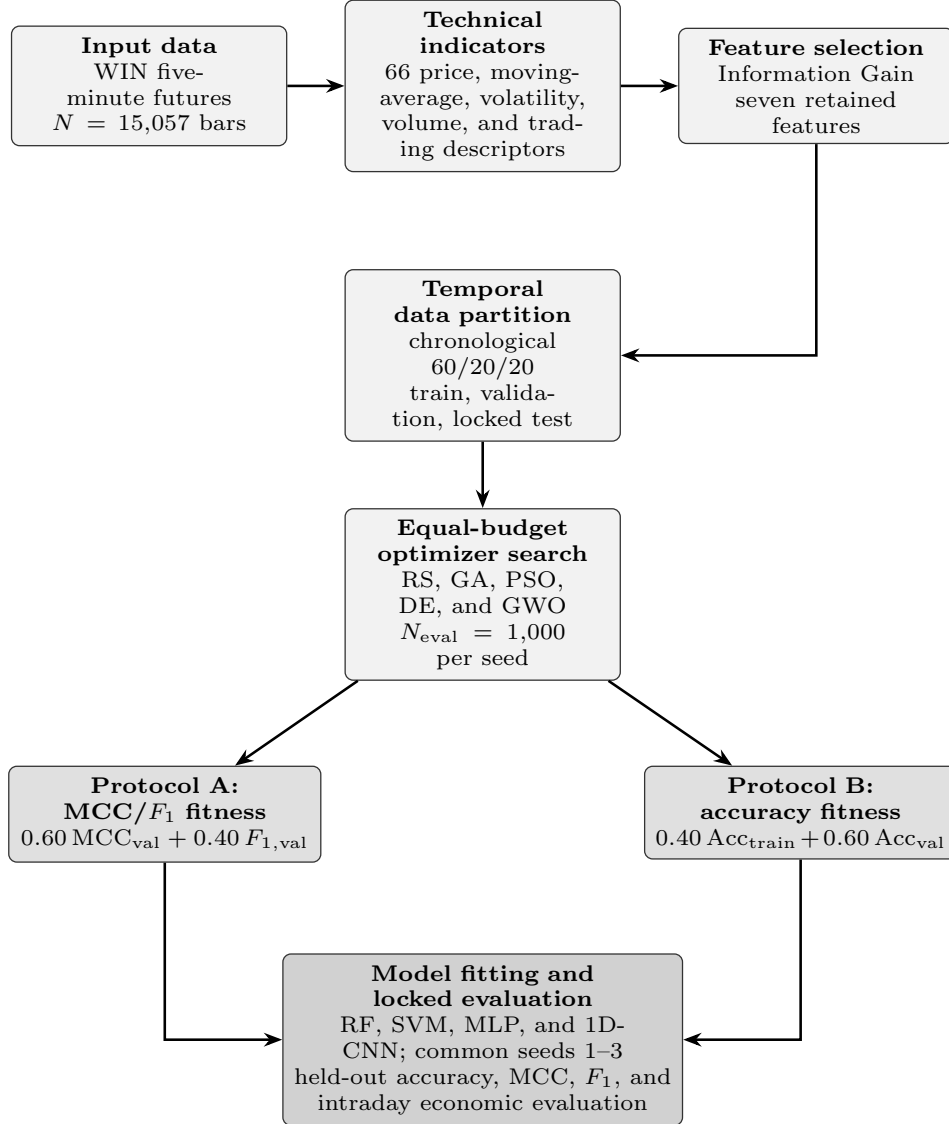
| Study & Year                                           | Domain & Data                               | Models                               | Optimization & Features                          | Validation Protocol                     | Methodological Gap                                                                          |
|--------------------------------------------------------|---------------------------------------------|--------------------------------------|--------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------|
| Ballings et al. (2015) [13]<br>Ding et al. (2011) [21] | Equities (5,767 firms)<br>Stock index trend | SVM, RF, AdaBoost<br>Neural Networks | Grid search / Manual<br>Hybrid GA-BP             | Cross-validation<br>Standard split      | Absence of metaheuristic search. Unequal evaluation budget across optimizers.               |
| Naik & Mohan (2019) [14]<br>Ecer et al. (2020) [15]    | Equities (NSE India)<br>Financial trend     | ANN classifier<br>MLP classifier     | Boruta feature selection<br>GA and PSO tuning    | Train/test split<br>K-fold CV           | Static parameters; no metaheuristics. Unconstrained budget; single classifier.              |
| Billah et al. (2024) [10]                              | Intraday equities                           | Heuristic models                     | Moving average rules                             | Standard split                          | Absence of neural hyperparameter tuning.                                                    |
| Bareket & Párv (2024) [33]                             | Surge detection                             | Adaptive ML                          | Technical indicators                             | Medium-term split                       | Evaluation restricted to static features.                                                   |
| Bandgar et al. (2025) [16]                             | Financial forecasting                       | Ensembles                            | Default tuning                                   | Random split                            | No evaluation budget cap; data leakage.                                                     |
| Rahim et al. (2025) [34]                               | 5-min futures                               | Single neural model                  | Single GA optimization                           | Train/test split                        | Single GA; no multi-swarm benchmark.                                                        |
| Dhingra et al. (2025) [9]                              | Solar PV forecasting                        | LSTM, CNN-LSTM                       | RAdam, Lookahead, Adam                           | Sequential split                        | Non-financial domain; stochastic optimizers omitted.                                        |
| Mutinda & Yong (2026) [30]                             | Financial market                            | Ensembles                            | SEDT feature selection                           | K-fold CV                               | Feature selection without tuning.                                                           |
| Simatupang et al. (2026) [35]                          | Stock index (IDX)                           | Bayesian RF                          | Bayesian optimization                            | Random K-fold CV                        | Random CV causes temporal data leakage.                                                     |
| Al Bannoud et al. (2026) [4]                           | Kinetics time series                        | ANN-ODE surrogate                    | GWO, PSO, GA, ABC                                | Multi-partition                         | Non-financial; unconstrained search budget.                                                 |
| Bashir et al. (2026) [1]<br>[Anonymized A] (2025) [3]  | Equities (NSE India)<br>Intraday B3 futures | Bagging Ensemble<br>Random Forest    | (1+1) Evolutionary Algo<br>InfoGain + Default RF | Train/test split<br>Sequential 60/20/20 | Single EA evaluated without budget cap. Untuned baseline classifier without metaheuristics. |
| [Anonymized B] (2026) [36]                             | Intraday B3 futures                         | MLP, SVM, RF                         | InfoGain + Single GA                             | 5-fold CV                               | Single GA; unconstrained 1,000-generation budget.                                           |
| <b>Present Work</b>                                    | <b>Intraday B3 futures</b>                  | <b>MLP, SVM, RF, CNN</b>             | <b>InfoGain + RS, GA, PSO, DE, GWO</b>           | <b>Sequential 60/20/20</b>              | <b>Equal budget</b> ( $N_{eval} = 1,000$ ), <b>3 seeds, fee backtest.</b>                   |

### 3 Methodology

This section presents the methodology adopted in this study. Figure 1 provides an overview of the flow from data acquisition to final model evaluation. It preserves the practical sequence of the IJCNN precursor—input data, technical indicators, feature selection, optimization, model fitting, and evaluation—while inserting the temporal controls and the two validation objectives required by the expanded benchmark.

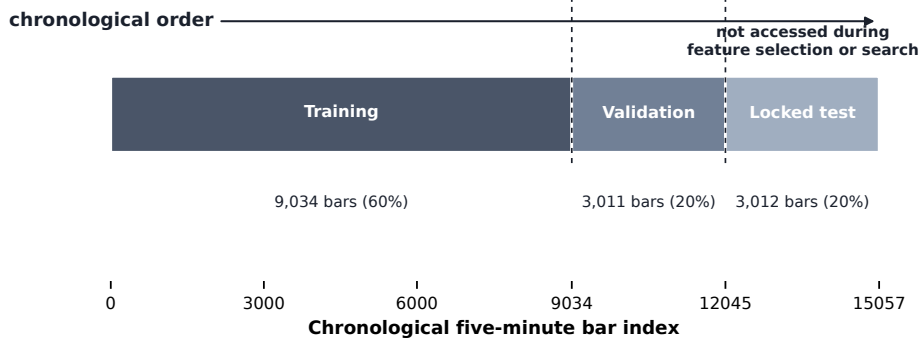
#### 3.1 Input Data

The data were obtained from the BovDB resources [6, 7]. The analyzed subset contains 15,057 clean five-minute records of the WIN contract from January to June 2024. WIN is referenced to the Ibovespa index and was selected because its liquidity and frequent intraday movement make it suitable for short-horizon computational prediction. The binary target indicates the next-bar direction: 0 for downtrend and 1 for uptrend.



**Fig. 1** Overview of the proposed methodology. The new benchmark preserves the precursor workflow while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

The records remain in chronological order. The first 9,034 observations are used for training, the subsequent 3,011 observations for validation, and the final 3,012 observations for locked testing. No random shuffling is applied. Figure 2 makes this separation explicit; the test observations are unavailable during both feature and hyperparameter selection.



**Fig. 2** Chronological data partition used in every experiment. Training, validation, and locked test segments occupy consecutive periods of the WIN five-minute series.

### 3.2 Generation of Technical Analysis Indicators

Technical indicators were derived from open, high, low, close, trade volume, and transaction-frequency information. Simple Moving Average (SMA), Exponential Moving Average (EMA), and dispersion measures were computed over short windows (3, 5, 7, and 9 periods) to retain sensitivity to intraday changes. For example, the difference between the seven- and three-period SMA is

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3. \quad (1)$$

The raw and derived variables were normalized before model fitting. Together, the construction procedure produces 66 descriptors, including moving-average differences and normalized counterparts.

### 3.3 Feature Selection Method

The indicator pool is reduced using Information Gain, as in the precursor. This filter selects attributes that most reduce uncertainty regarding the next-bar direction and avoids repeating highly redundant descriptors in each search evaluation [19, 20]. Seven features are retained:  $\text{EMA}_{5-3}$ , normalized  $\text{EMA}_{5-3}$ ,  $\text{EMA}_{7-3}$ , normalized  $\text{EMA}_{7-3}$ , normalized  $\text{SMA}_9$ ,  $\text{EMA}_{9-3}$ , and normalized  $\text{EMA}_7$ . The same reduced representation is used by all model and optimizer combinations.

### 3.4 Optimizer Configuration and Validation Objectives

Five optimizers are evaluated: RS, GA, PSO, DE, and GWO. Each optimizer receives 1,000 candidate evaluations for each model and seed. Population-based methods use populations of ten candidates, corresponding to 100 generations or iterations; RS samples 1,000 feasible configurations. GA uses tournament selection, crossover, and mutation; PSO uses inertia plus cognitive and social terms; DE uses the best/1/bin strategy; and GWO updates candidate positions from its leading agents.

Two validation objectives are evaluated. Protocol A uses the compound metric

$$f_{\text{MCC/F1}}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}), \quad (2)$$

whereas Protocol B uses

$$f_{\text{Acc}}(\boldsymbol{\theta}) = 0.40 \text{Accuracy}_{\text{train}}(\boldsymbol{\theta}) + 0.60 \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (3)$$

After candidate selection, the selected configuration is refit on training plus validation data and evaluated once on the locked test block.

## 4 Results

This section presents the experimental results and their implications for short-term prediction of Brazilian Mini-Index futures. Following the organization of the precursor, it first describes the common experimental setup and then discusses the performance of the individual model families. The expanded benchmark subsequently reports the direct cross-model comparison, exploratory statistical evidence, and economic outcomes.

### 4.1 Experimental Setup and Data Partitioning

All model–optimizer combinations use the seven descriptors described in Section 3.3, the chronological split shown in Figure 2, and the same 1,000-evaluation allowance per seed. RF, SVM, MLP, and 1D-CNN each are searched with RS, GA, PSO, DE, and GWO. The two protocols differ only in the validation fitness used to rank candidate configurations. Three common seeds (1–3) are available for every cell in both protocols.

Consequently, the direct comparison between fitness functions is descriptive: each reported model–optimizer result is the mean over the same three seeds. The locked test block contains 3,012 observations and is not used during search. This design differs from the randomized training and test sampling used in the precursor and removes look-ahead exposure from model selection.

### 4.2 Performance of the Models

Tables 2–5 retain the presentation pattern of the precursor: each table states the search space and an illustrative GA-selected configuration. The configurations are from seed 1 under Protocol A and are included to make the optimization procedure concrete;

**Table 2** Random Forest hyperparameters and representative GA-selected configuration.

| Hyperparameter          | Search space       | Selected value |
|-------------------------|--------------------|----------------|
| $n_{\text{estimators}}$ | [50, 500]          | 58             |
| max_depth               | [5, 50]            | 6              |
| min_samples_split       | [2, 20]            | 13             |
| min_samples_leaf        | [1, 20]            | 15             |
| max_features            | {sqrt, log2, None} | None           |
| class_weight            | {None, balanced}   | None           |

**Table 3** SVM hyperparameters and representative GA-selected configuration.

| Hyperparameter | Search space                  | Selected value |
|----------------|-------------------------------|----------------|
| $C$            | $[10^{-3}, 10^3]$ (log scale) | 2.089          |
| $\gamma$       | $[10^{-5}, 10^1]$ (log scale) | 5.044          |
| Kernel         | {linear, RBF}                 | linear         |

**Table 4** MLP hyperparameters and representative GA-selected configuration.

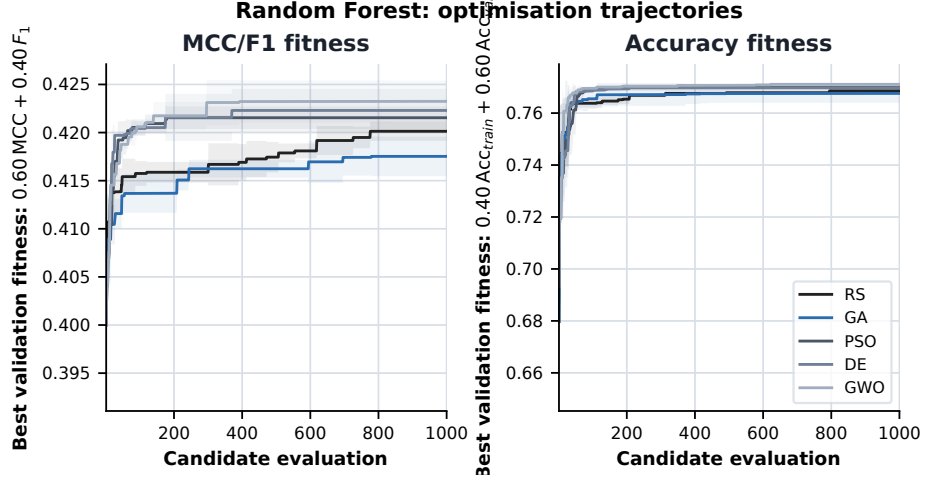
| Hyperparameter       | Search space                     | Selected value        |
|----------------------|----------------------------------|-----------------------|
| Hidden neurons       | [5, 500]                         | 410                   |
| Learning rate        | $[10^{-8}, 10^{-2}]$ (log scale) | $3.42 \times 10^{-4}$ |
| $L_2$ regularization | $[10^{-8}, 10^{-2}]$ (log scale) | $1.66 \times 10^{-7}$ |
| Dropout rate         | [0, 0.10]                        | 0.057                 |
| Batch size           | {16, 32, 64, 80, 112, 128}       | 112                   |

**Table 5** 1D-CNN hyperparameters and representative GA-selected configuration.

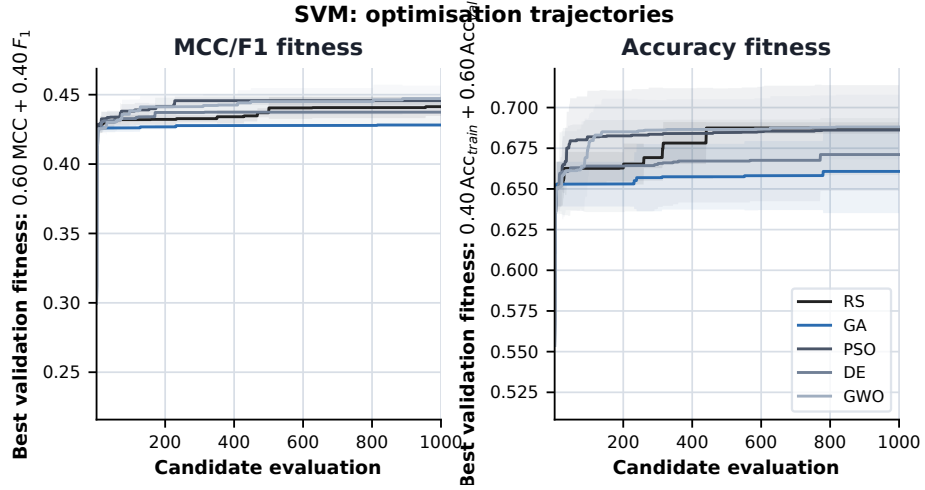
| Hyperparameter       | Search space                     | Selected value        |
|----------------------|----------------------------------|-----------------------|
| Filters              | [8, 128]                         | 48                    |
| Kernel size          | [2, 5]                           | 3                     |
| Dense neurons        | [16, 256]                        | 251                   |
| Learning rate        | $[10^{-8}, 10^{-2}]$ (log scale) | $3.90 \times 10^{-6}$ |
| $L_2$ regularization | $[10^{-8}, 10^{-2}]$ (log scale) | $5.70 \times 10^{-6}$ |
| Dropout rate         | [0, 0.10]                        | 0.026                 |
| Batch size           | {16, 32, 64, 80, 112, 128}       | 16                    |

they are not asserted to be universally optimal because all optimizer–seed outcomes are summarized below.

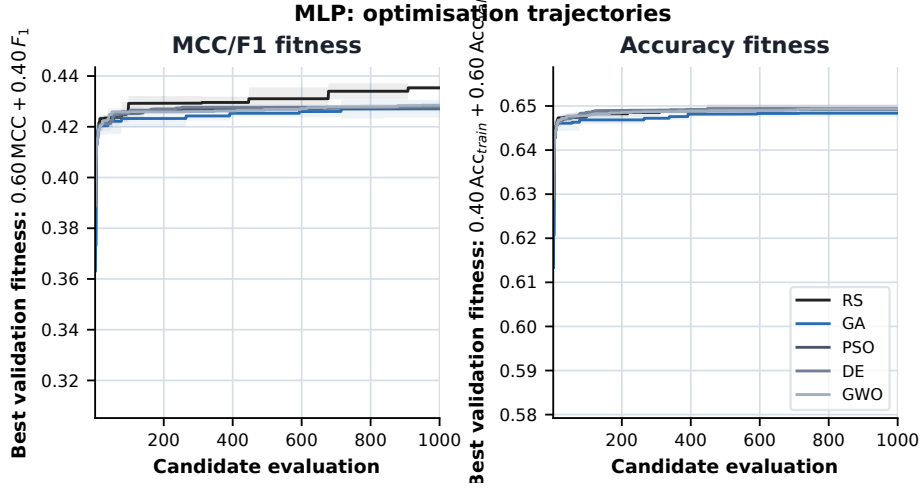
Figures 3–6 replace the three GA-only convergence plots of the precursor with the five budget-matched search trajectories. For each model, the left panel follows the compound  $\text{MCC}/F_1$  fitness and the right panel follows weighted training/validation accuracy. The vertical scales differ because the objectives differ; their role is to show search convergence within a protocol, not to compare objective magnitudes across protocols.



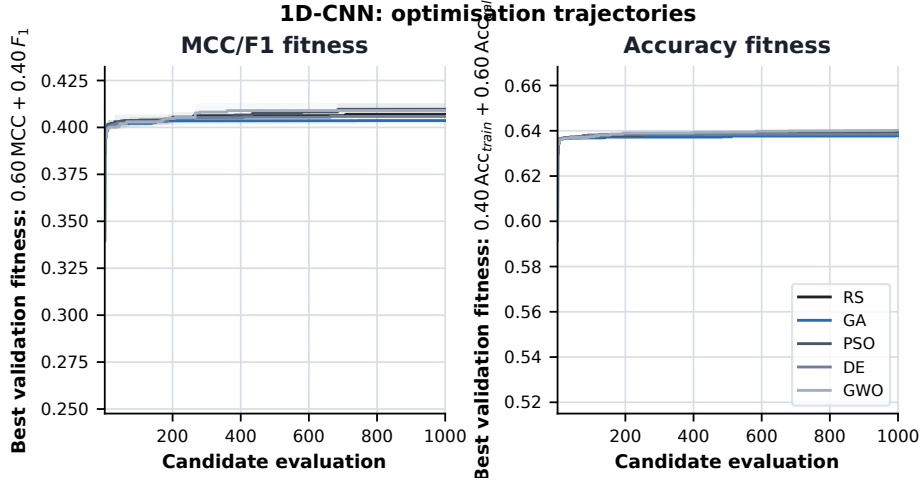
**Fig. 3** Fitness evolution during equal-budget optimization of the Random Forest model. Lines summarize the three common seeds.



**Fig. 4** Fitness evolution during equal-budget optimization of the SVM model. Lines summarize the three common seeds.



**Fig. 5** Fitness evolution during equal-budget optimization of the MLP model. Lines summarize the three common seeds.

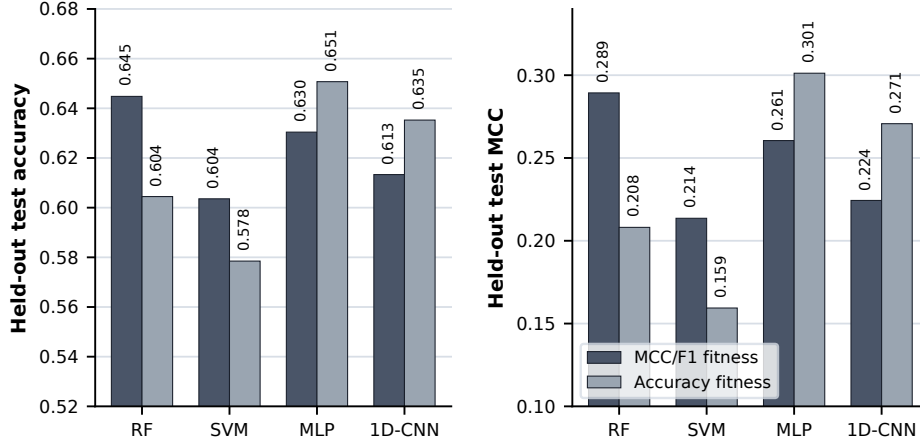


**Fig. 6** Fitness evolution during equal-budget optimization of the 1D-CNN model. Lines summarize the three common seeds.

Table 6 and Figure 7 present held-out performance aggregated over 15 optimizer–seed blocks per model within each protocol. Under  $\text{MCC}/F_1$  fitness, RF has the largest mean test accuracy (0.645), MCC (0.289), and  $F_1$  (0.664). Under weighted accuracy fitness, MLP has the largest mean test accuracy (0.651), MCC (0.301), and  $F_1$  (0.658). The reversal reinforces that the validation criterion and model family should be reported separately rather than treated as interchangeable.

**Table 6** Mean locked-test predictive outcomes by backbone and validation protocol (15 optimizer–seed blocks per row).

| Model  | MCC/ $F_1$ fitness |       |       | Weighted accuracy fitness |       |       |
|--------|--------------------|-------|-------|---------------------------|-------|-------|
|        | Accuracy           | MCC   | $F_1$ | Accuracy                  | MCC   | $F_1$ |
| RF     | 0.645              | 0.289 | 0.664 | 0.604                     | 0.208 | 0.624 |
| SVM    | 0.604              | 0.214 | 0.602 | 0.578                     | 0.159 | 0.600 |
| MLP    | 0.630              | 0.261 | 0.639 | 0.651                     | 0.301 | 0.658 |
| 1D-CNN | 0.613              | 0.224 | 0.597 | 0.635                     | 0.271 | 0.639 |

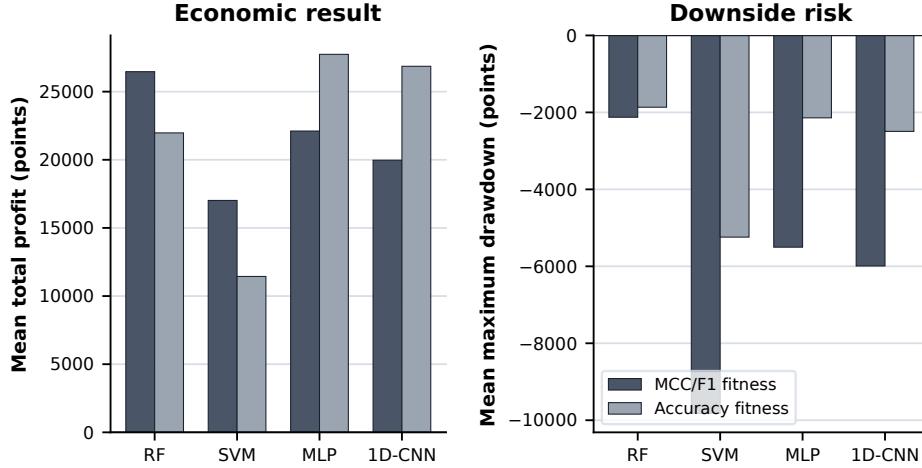


**Fig. 7** Comparison of mean locked-test accuracy and MCC for all four model families under the two validation objectives.

### 4.3 Exploratory Statistical and Economic Evaluation

The model-level tests are performed within each holdout protocol. The five optimizer–seed combinations yield 15 matched blocks for each backbone. Under MCC/ $F_1$  fitness, the Friedman comparison of test MCC is significant ( $\chi^2 = 22.84$ ,  $p = 4.36 \times 10^{-5}$ ), with RF obtaining the best mean outcome. Under weighted accuracy fitness, the comparison of test accuracy is significant ( $\chi^2 = 42.92$ ,  $p = 2.56 \times 10^{-9}$ ), with MLP first; its Holm-adjusted pairwise contrasts against RF, SVM, and 1D-CNN are significant ( $p_{\text{Holm}} \approx 0.00194$ ). These tests compare models within a protocol and should not be interpreted as a confirmatory test between the two fitness functions.

The economic evaluation yields a complementary ranking. Under MCC/ $F_1$  fitness, RF has the largest mean total profit (26,465 points). Under weighted accuracy fitness, MLP has the largest mean total profit (27,737 points), whereas RF has the least negative mean maximum drawdown (−1,866 points). Figure 8 therefore makes explicit the difference between predictive ranking and return-risk ranking.



**Fig. 8** Mean total profit and maximum drawdown across 15 optimizer–seed blocks per model. Results are reported separately for the two validation objectives.

## 5 Conclusion

This work extends the IJCNN precursor on technical indicators, Information Gain feature selection, and GA-based hyperparameter optimization for Mini-Index futures prediction. The original practical pipeline is retained: a compact set of technical descriptors is constructed from intraday WIN data, informative attributes are selected, and nonlinear models are optimized before final testing. The present study expands that design through a chronological train/validation/test split, four model families, five budget-matched optimizers, two explicit validation objectives, and a locked economic evaluation.

The results show that a single validation score cannot define a universal model ranking. RF produces the strongest mean locked-test accuracy, MCC, and  $F_1$  under the compound MCC/ $F_1$  fitness, whereas MLP leads the predictive results under weighted accuracy fitness. The economic analysis reaches a related but distinct conclusion: MLP has the largest mean profit under weighted accuracy fitness, while RF has the smallest mean drawdown. These findings support the use of controlled metaheuristic optimization, but they also show why the selection objective, reported predictive metric, and economic endpoint must be stated separately.

The direct comparison between fitness functions remains descriptive because only three seeds are common to every model–optimizer cell. The statistical analysis is correspondingly limited to exploratory within-protocol comparisons. Future work should complete a preregistered common 20-seed design, retain the same temporal split and evaluation budget, and repeat the analysis across additional liquid futures markets and market regimes. Such an extension would test the stability of the present results while preserving the transparent workflow established here.

## Declarations

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### Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

### Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

### Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

### Author Contributions

All authors contributed to the study conception and design. Data collection and experimental execution were performed by [Anonymous for double-blind review]. Statistical analysis and manuscript preparation were carried out by [Anonymous for double-blind review]. All authors read and approved the final manuscript.

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